

Challenge #1: Arabic/Darija Patient Pharmacy Assistance

Context

In healthcare-related digital services in Morocco, users often express their needs through short, incomplete, informal, or voice-based messages. This is particularly challenging in multilingual environments where users may alternate between Arabic, Darija, or French. In a patient-pharmacy relationship, this may be even more challenging. Patients could ask many questions about medications availability or pharmacy information (phone number, localisation, etc.). An intelligent conversational assistant can help interpret such requests and transform them into structured information that can be processed by downstream systems.

Project Description

The objective of this project is to design and implement an intelligent service, capable of assisting users in formulating medication-related requests to their pharmacies. The system should accept text and/or voice input, analyze the content, identify the user's intent, and extract relevant information such as medication names, quantities, or other useful details.

The proposed solution should support multilingual interaction, with particular attention to Arabic, Darija, and French, and should be able to return a structured interpretation of the request rather than only producing free-form responses. The project combines natural language processing, speech processing, and backend API development in a practical healthcare-oriented use case.

Main Objectives

- Design a conversational assistance system adapted to pharmacy-related requests
- Process user input in text and/or voice format
- Support multilingual or code-mixed interaction, especially Arabic, Darija, and French
- Extract useful structured information from unstructured requests
- Expose the final system through an API

Expected Output

- Study of the problem and related work
- Design of the chatbot architecture and processing pipeline
- Implementation of the NLP and/or speech processing components

- Development of an API service for interacting with the system
- Test scenarios and evaluation of the obtained results

Challenge #2: Medication Box Recognition Using Computer Vision and OCR

Context

Recognizing a medication from an image of its box or packaging can be highly useful in digital healthcare applications. Such recognition is challenging because it may involve visual similarity between products, varying image quality, and multilingual text appearing on packaging, especially in Arabic and French. Reliable identification therefore requires combining image analysis with text extraction and matching techniques.

Project Description

This project aims to develop an intelligent service capable of recognizing a medication from a user-provided image. The system should analyze the visual content of the image, extract relevant textual information using OCR, and compare the obtained results with a reference medication database in order to identify the most likely product.

Particular attention should be given to multilingual packaging and to the robustness of the recognition pipeline under realistic conditions such as blur, lighting variation, or partial visibility of the box. The final solution should be delivered as an API that can be used by external applications. The project combines computer vision, OCR, text matching, and backend service development.

Main Objectives

- Study the problem of medication identification from packaging images
- Develop a computer vision and OCR pipeline for recognition
- Handle Arabic and French text appearing on medication boxes
- Match extracted information against a medication reference database
- Expose the final recognition system through an API

Expected output

- Review of methods related to OCR and image-based recognition
- Design of the proposed recognition pipeline
- Construction or preparation of a medication image dataset
- Implementation of the recognition and matching system
- Development of an API for querying the service

- Experimental evaluation and analysis of system performance